

PRAKTIKUM SISTEM OPERASI

**MODUL 15
(KEAMANAN)**



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

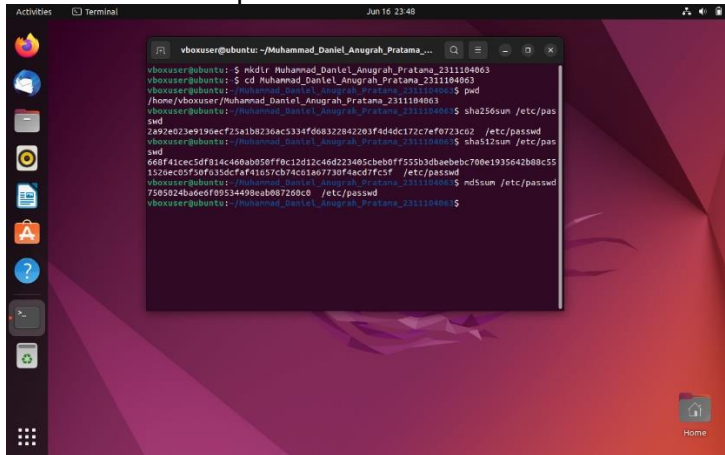
UNIVERSITAS TELKOM

2026

JURNAL/TP

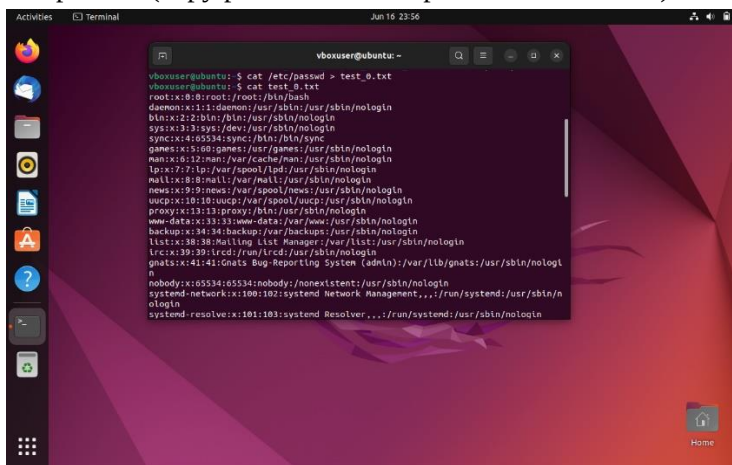
1. Integritas: dasar hashing

a. Lakukan hash SHA256, SHA512 dan MD5 untuk file /etc/passwd. Berapa nilai hash dari file /etc/passwd? Screenshot nilai hash dari file tersebut.



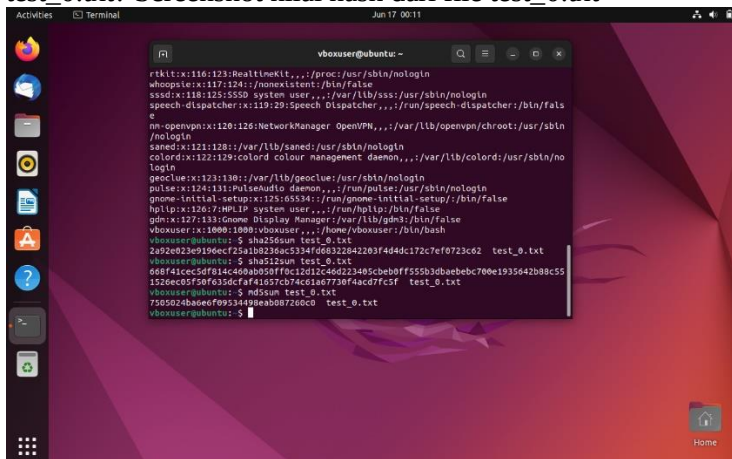
```
vboxuser@ubuntu: ~$ md5sum /etc/passwd
2a92e023e9190ecf25a1b236ac5334f068322842203f4d4dc172c7ef0723c62 /etc/passwd
vboxuser@ubuntu: ~$ sha256sum /etc/passwd
668f41cec5df814c46a8ab050ff0c12d12c46d223405cbe0ff555b3dbaebebc70e1935642b8c55
1526ec05f50f3d5cfaf41657cb74c0a67730f4acd7fc5f /etc/passwd
vboxuser@ubuntu: ~$ sha512sum /etc/passwd
7505024ba6ef0953449ea0807206c0 /etc/passwd
```

b. Buatlah file bernama test_0.txt pada folder /home/praktikan. Isi file tersebut isi yang ada di file /etc/passwd (copy paste isi file /etc/passwd ke test_0.txt)



```
vboxuser@ubuntu: ~$ cat /etc/passwd > test_0.txt
vboxuser@ubuntu: ~$ cat test_0.txt
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:12:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:11:11:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
lirc:x:39:39:lirc:/run/lirc:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin)/:/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:100:100:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:101:101:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
```

c. Lakukan hash SHA256, SHA512 dan MD5 untuk file test_0.txt. Berapa nilai hash dari file test_0.txt? Screenshot nilai hash dari file test_0.txt



```
rtkit:x:116:123:RealtimeKit,,,/proc:/usr/sbin/nologin
whoopsie:x:117:124:nonexistent:/bin/false
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
nn-ovpn:x:120:126:NetworkManager OpenVPN,,,:/var/lib/ovpn/chroot:/usr/sbin/nologin
saned:x:121:128::/var/lib/saned:/usr/sbin/nologin
colord:x:122:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:123:130::/var/lib/geoclue:/usr/sbin/nologin
pulse:x:124:131:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:125:65534::/run/gnome-initial-setup:/bin/false
hplip:x:136:7:HP Linux system user,,,:/run/hplip:/bin/false
gdm:x:127:133:Gnome Display Manager:/var/lib/gdm:/bin/false
vboxuser:x:1000:1000:vboxuser,,,:/home/vboxuser:/bin/bash
vboxuser@ubuntu: ~$ sha256sum test_0.txt
2a92e023e9190ecf25a1b236ac5334f068322842203f4d4dc172c7ef0723c62 test_0.txt
vboxuser@ubuntu: ~$ sha512sum test_0.txt
668f41cec5df814c46a8ab050ff0c12d12c46d223405cbe0ff555b3dbaebebc70e1935642b8c55 test_0.txt
vboxuser@ubuntu: ~$ md5sum test_0.txt
7505024ba6ef0953449ea0807206c0 test_0.txt
vboxuser@ubuntu: ~$
```

e. Apa hasil pengamatan Anda? File apa saja yang mempunyai hash yang sama? Jelaskan!

[illegible]

The screenshot shows a Linux desktop with a purple background. On the left is a dock with icons for Activities, Terminal, Firefox, and several other applications. The top panel displays 'Activities', 'Terminal', and the date 'Jun 17 00:21'. The main window is a terminal titled 'vboxuser@ubuntu: ~' with the following content:

```
vboxuser@ubuntu: ~
bKSfTelkomuniversi7sFackSfLn2FDocument4N2FfLABN20P8rIOEN2025N2026N2FCAN2ILK
ZfPRAKtIKUMN2fBANKN26S0ALN2F51SOPKZfMODULN2014N20N2D26KEAPANNAN20LINUKS2FFILN2K
000NULADAgaw1
Reusing existing connection to telkomuniversi7official-my.sharepoint.com:443.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'test1.txt'

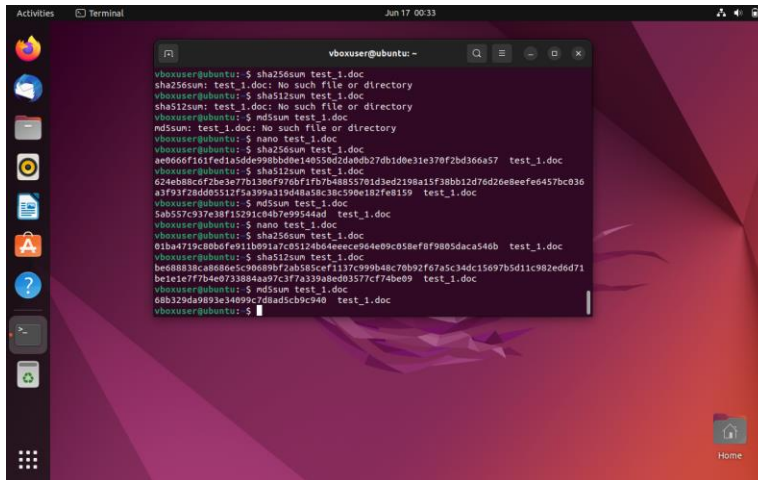
test1.txt           [ <<> ] 386.84K 1.59MB/s   in 0.2s

2026-00-17 00:18:25 (1.59 MB/s) - 'test1.txt' saved [396129]

vboxuser@ubuntu:~$ ls
Backup      file_0.txt
Documents   Muhammad_sental_Awagrah_Prataha_231104083  Pictures  Templates
Downloads   Mark                                         Snap      test1.txt
vboxuser@ubuntu:~$ sha256sum test.1.txt
sha256sum: test.1.txt: No such file or directory
vboxuser@ubuntu:~$ sha512sum test.1.txt
sha512sum: test.1.txt: No such file or directory
vboxuser@ubuntu:~$ md5sum test.1.txt
md5sum: test.1.txt: No such file or directory
vboxuser@ubuntu:~$
```

The terminal output shows a successful HTTP connection to a SharePoint site, followed by a file download of 'test1.txt'. The file size is 386.84K and it was downloaded at 1.59MB/s. The file was saved successfully. The user then lists the contents of the home directory, showing files like 'Backup', 'file_0.txt', 'Documents', 'Downloads', 'Mark', 'Pictures', 'Templates', 'Snap', and 'test1.txt'. Finally, the user runs several checksum commands (sha256sum, sha512sum, md5sum) on 'test.1.txt', which all fail with the message 'No such file or directory'.

JAWABAN UNTUK 3b-e



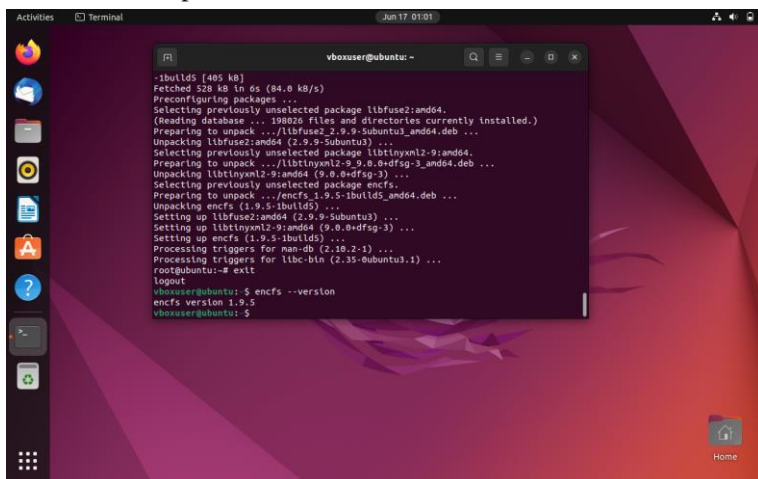
```
vboxuser@ubuntu:~$ sha256sum test_1.doc
sha256sum: test_1.doc: No such file or directory
vboxuser@ubuntu:~$ sha512sum test_1.doc
sha512sum: test_1.doc: No such file or directory
vboxuser@ubuntu:~$ md5sum test_1.doc
md5sum: test_1.doc: No such file or directory
vboxuser@ubuntu:~$ nano test_1.doc
vboxuser@ubuntu:~$ sha256sum test_1.doc
aef6d6f101feda154e99ab0de140550d2da0b27db1d0e31e378f2bd366a57 test_1.doc
vboxuser@ubuntu:~$ sha512sum test_1.doc
624eb88cf2be3e77b1306f976bf7b7b48855701d3ed2198a15f38bb12d76d2e0eef6e457bc036
a1f9172dd0512f7a399a31d404ad3c39ee182fe0139 test_1.doc
vboxuser@ubuntu:~$ md5sum test_1.doc
5ab557c937e38f15291c84b7e99544ad test_1.doc
vboxuser@ubuntu:~$ nano test_1.doc
vboxuser@ubuntu:~$ sha256sum test_1.doc
01ba4719c88b0fe911b091a7c85124b4eece964e09c058ef8f9805daca540b test_1.doc
vboxuser@ubuntu:~$ sha512sum test_1.doc
be688038ca868e5c90689b72ab585cef1137c999b48c70b92f67a5c34dc15697b5d11c982eded71
be1e1e777b4e0733884aa7c377a339a8e003577cf74be09 test_1.doc
vboxuser@ubuntu:~$ md5sum test_1.doc
68b329da9893e34099c7daad5cb9c940 test_1.doc
vboxuser@ubuntu:~$
```

4. Konfidensialitas: encfs

a. Jalankan perintah berikut:

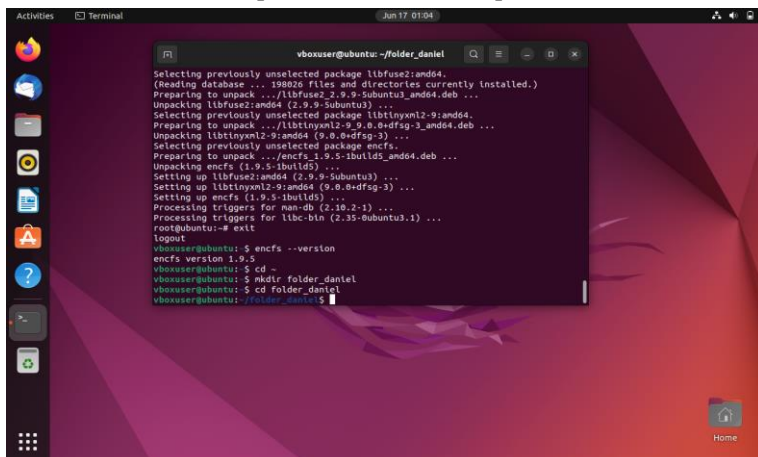
`encfs ~/folder_anda/folder_terenkripsi ~/folder_anda/folder_normal`

Pilih y (enter), pilih y (enter), enter, kemudian buatlah password. Ingatlah password yang dibuat. Setelah selesai, perhatikan bahwa telah muncul folder bernama `folder_normal` dan `folder_terenkripsi`.



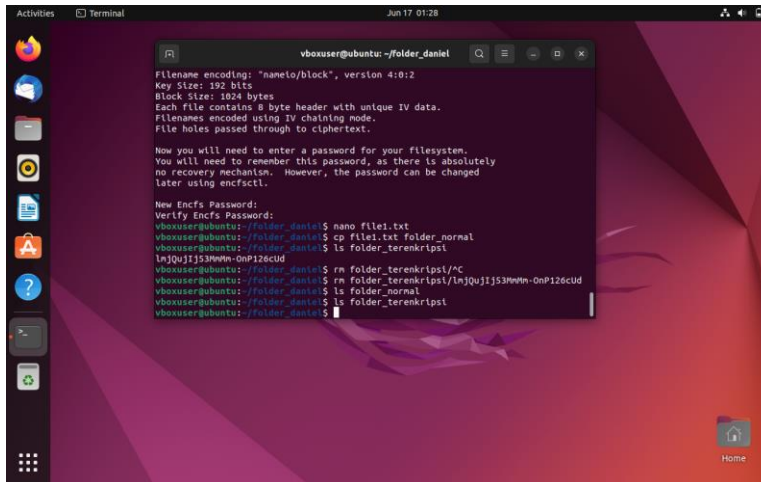
```
-ibuilds [aps kb]
Fetched 528 kB in 0s (84.0 kB/s)
Preconfiguring packages ...
Selecting previously unselected package libfuse2:amd64.
(Reading database ... 198026 files and directories currently installed.)
Preparing to unpack .../libfuse2_2.9.9-Sububuntu.1_amd64.deb ...
Unpacking libfuse2:amd64 (2.9.9-Sububuntu.1) ...
Selecting previously unselected package libtinyinml2-9:amd64.
Preparing to unpack .../libtinyinml2-9.9.0-0dfsg-3_amd64.deb ...
Unpacking libtinyinml2-9:amd64 (9.9.0-0dfsg-3) ...
Selecting previously unselected package encfs.
Preparing to unpack .../encfs_1.9.5-1build5_amd64.deb ...
Unpacking encfs (1.9.5-1build5) ...
Setting up libfuse2:amd64 (2.9.9-Sububuntu.1) ...
Setting up libtinyinml2-9:amd64 (9.9.0-0dfsg-3) ...
Setting up encfs (1.9.5-1build5) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu1.1) ...
root@ubuntu:~# exit
vboxuser@ubuntu:~$ encfs --version
encfs version 1.9.5
vboxuser@ubuntu:~$
```

b. Copy dua atau tiga buah file (file apa saja) ke `folder_normal`. Amati dan tulis hasil observasi Anda pada `folder_terenkripsi`!



```
vboxuser@ubuntu:~$ cp -r folder_normal folder_daniel
vboxuser@ubuntu:~$ cd ~
vboxuser@ubuntu:~$ ls -l folder_daniel
total 4
drwxr-xr-x 2 vboxuser vboxuser 4096 Jun 17 01:04 .
drwxr-xr-x 3 vboxuser vboxuser 4096 Jun 17 01:04 ..
-rw-r--r-- 1 vboxuser vboxuser 123456 Jun 17 01:04 file1.txt
-rw-r--r-- 1 vboxuser vboxuser 123456 Jun 17 01:04 file2.txt
vboxuser@ubuntu:~$
```


c. Hapus salah satu file (bebas) pada folder_terenkripsi. Amati dan tulis hasil observasi Anda pada folder_normal!



```
vboxuser@ubuntu: ~/folder_daniel
Filename encoding: "nameio/block", version 4:0:2
Key Size: 192 bits
Block Size: 1024 bytes
Each file contains 8 byte header with unique IV data.
Filename encoding using IV chaining mode.
File holes passed through to ciphertext.

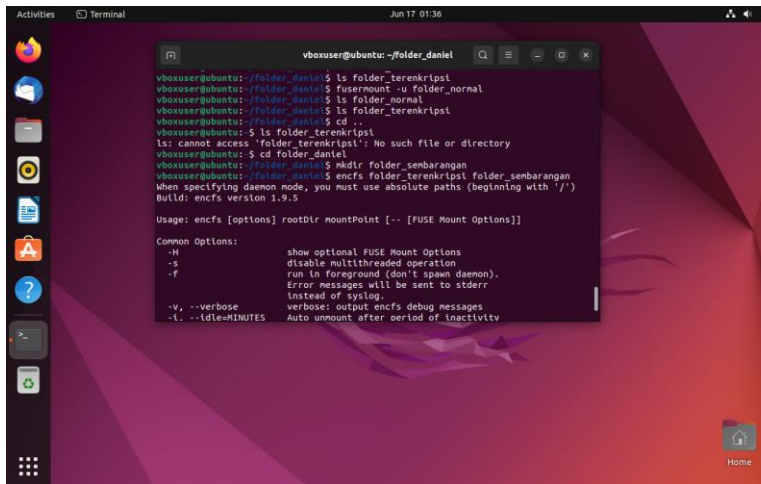
Now you will need to enter a password for your filesystem.
You will need to remember this password, as there is absolutely
no recovery mechanism. However, the password can be changed
later using encfsctl.

New Encfs Password:
Verify Encfs Password:
vboxuser@ubuntu: ~/folder_daniel$ nano file1.txt
vboxuser@ubuntu: ~/folder_daniel$ cp file1.txt folder_normal
vboxuser@ubuntu: ~/folder_daniel$ ls folder_terenkripsi
lnJQuJl53MMm-OnP126cud
vboxuser@ubuntu: ~/folder_daniel$ rm folder_terenkripsi/*C
vboxuser@ubuntu: ~/folder_daniel$ rm folder_terenkripsi/lnJQuJl53MMm-OnP126cud
vboxuser@ubuntu: ~/folder_daniel$ ls folder_normal
vboxuser@ubuntu: ~/folder_daniel$ ls folder_terenkripsi
```

d. Lakukan umount dengan perintah:

Fusermount -u ~/folder_anda/folder_normal

Amati dan tulis hasil observasi Anda pada folder_normal dan folder_terenkripsi



```
vboxuser@ubuntu: ~/folder_daniel
vboxuser@ubuntu: ~/folder_daniel$ ls folder_terenkripsi
vboxuser@ubuntu: ~/folder_daniel$ fusermount -u folder_normal
vboxuser@ubuntu: ~/folder_daniel$ ls folder_normal
vboxuser@ubuntu: ~/folder_daniel$ ls folder_terenkripsi
vboxuser@ubuntu: ~/folder_daniel$ cd ..
vboxuser@ubuntu: $ ls folder_terenkripsi
ls: cannot access 'folder_terenkripsi': No such file or directory
vboxuser@ubuntu: $ cd folder_daniel
vboxuser@ubuntu: ~/folder_daniel$ mkdir folder_senbarang
vboxuser@ubuntu: ~/folder_daniel$ encfs folder_terenkripsi folder_senbarang
When specifying daemon mode, you must use absolute paths (beginning with '/')
Build: encfs version 1.9.5

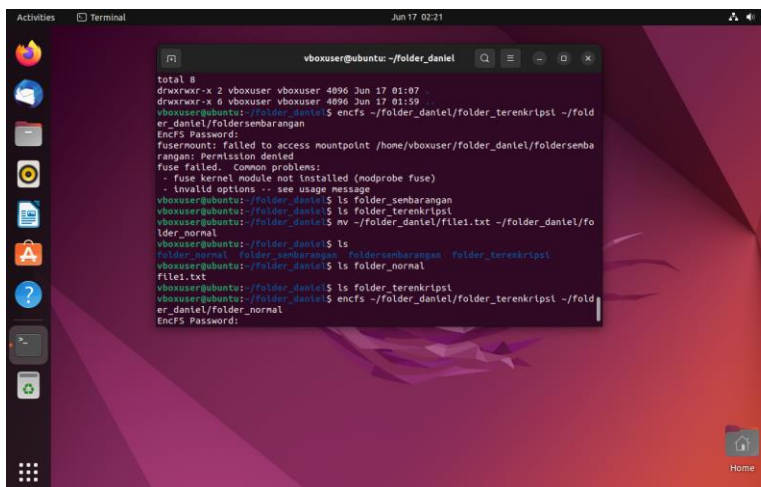
Usage: encfs [options] rootDir mountPoint [-- [FUSE Mount Options]]

Common Options:
  -H          show optional FUSE Mount Options
  -s          disable multithreaded operation
  -f          run in foreground (don't spawn daemon),
             Error messages will be sent to stderr
             instead of syslog.
  -v, --verbose verbose: output encfs debug messages
  -i, --idle=MINUTES Auto unmount after period of inactivity
```

e. Buatlah folder baru bernama folder_senbarang. Lakukan perintah berikut ini:

encfs ~/folder_anda/folder_terenkripsi ~/folder_anda/folder_senbarang

Amati dan tulis hasil observasi Anda!



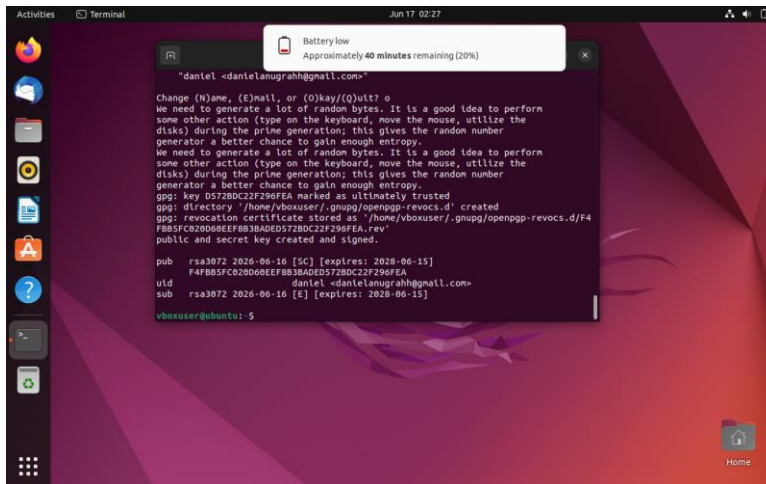
```
vboxuser@ubuntu: ~/folder_daniel
total 8
drwxr-xr-x 2 vboxuser vboxuser 4096 Jun 17 01:07
drwxr-xr-x 6 vboxuser vboxuser 4096 Jun 17 01:59
vboxuser@ubuntu: ~/folder_daniel$ encfs ~/folder_daniel/folder_terenkripsi ~/folder_daniel/folder_senbarang
Encfs Password:
fusermount: failed to access mountpoint /home/vboxuser/folder_daniel/folder_senbarang: Permission denied
fuse failed. Common problems:
- fuse kernel module not installed (modprobe fuse)
- invalid options -- see usage message
vboxuser@ubuntu: ~/folder_daniel$ ls folder_senbarang
vboxuser@ubuntu: ~/folder_daniel$ ls folder_terenkripsi
vboxuser@ubuntu: ~/folder_daniel$ mv ~/folder_daniel/file1.txt ~/folder_daniel/folder_normal
vboxuser@ubuntu: ~/folder_daniel$ ls
vboxuser@ubuntu: ~/folder_daniel$ ls folder_senbarang folder_terenkripsi
vboxuser@ubuntu: ~/folder_daniel$ ls
file1.txt
vboxuser@ubuntu: ~/folder_daniel$ ls folder_terenkripsi
vboxuser@ubuntu: ~/folder_daniel$ encfs ~/folder_daniel/folder_terenkripsi ~/folder_daniel/folder_normal
Encfs Password:
```

5. Konfidensialitas: gpg

a. Membuat kunci publik dan privat. Jalankan perintah ini:

`gpg --gen-key`

Baca dan ikuti perintah yang ada dari program gpg!



```
Activities Terminal Jun 17 02:27
Battery low
Approximately 40 minutes remaining (20%)

"daniel <danielnugrah@gmail.com>"

Change (N)ame, (E)mail, or (O)kay/(Q)uit? o
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: key 0572BDC22F296FEA marked as ultimately trusted
gpg: directory '/home/vboxuser/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/vboxuser/.gnupg/openpgp-revocs.d/F4F8B5FC02B060EEF8B3BADE572BDC22F296FEA.rev'
public and secret key created and signed.

pub rsa3072 2026-06-16 [SC] [expires: 2028-06-15]
F4F8B5FC02B060EEF8B3BADE572BDC22F296FEA
uid
sub rsa3072 2026-06-16 [E] [expires: 2028-06-15]

vboxuser@ubuntu: $
```

b. Hasil publickey dan privatekey ada di folder `~/.gnupg`. Privatekey tidak boleh keluar dari komputer ini dan hanya Anda saja yang dapat mengaksesnya. Kunci publik akan dibagikan kepada seluruh dunia atau untuk orang yang Anda inginkan saja. Kunci publik adalah `pubring.gpg` dan kunci private adalah `secring.gpg`

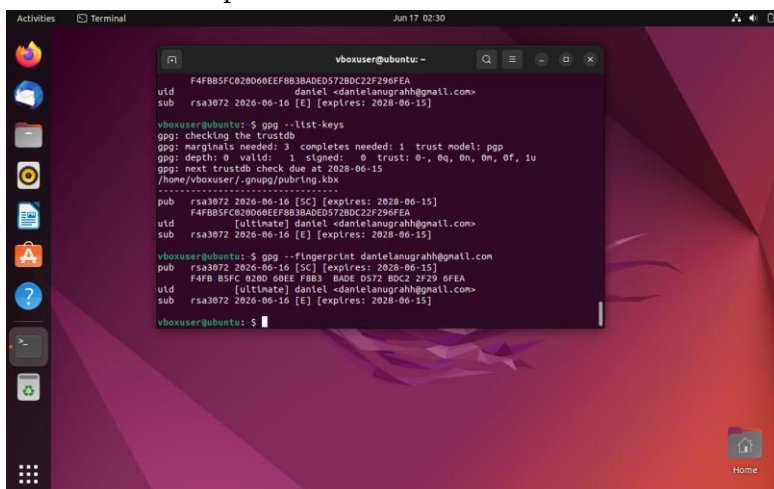
Lakukan perintah berikut ini untuk mengetahui daftar key yang Anda punya dan fingerprint

kunci yang Anda punya!

`gpg --list-keys`

`gpg --fingerprint nama@email.com`

Screenshot hasil perintah di atas!



```
Activities Terminal Jun 17 02:30

vboxuser@ubuntu: ~
F4F8B5FC02B060EEF8B3BADE572BDC22F296FEA
uid daniel <danielnugrah@gmail.com>
sub rsa3072 2026-06-16 [E] [expires: 2028-06-15]

vboxuser@ubuntu: $ gpg --list-keys
gpg: checking the trustdb
gpg: marginal: needed: 3, completes needed: 1, trust model: pgp
gpg: depth: 0, valid: 1, signed: 0, trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-06-15
/home/vboxuser/.gnupg/pubring.kbx
-----
pub rsa3072 2026-06-16 [SC] [expires: 2028-06-15]
F4F8B5FC02B060EEF8B3BADE572BDC22F296FEA
uid [ultimate] daniel <danielnugrah@gmail.com>
sub rsa3072 2026-06-16 [E] [expires: 2028-06-15]

vboxuser@ubuntu: $ gpg --fingerprint danielnugrah@gmail.com
pub rsa3072 2026-06-16 [SC] [expires: 2028-06-15]
F4F8B5FC02B060EEF8B3BADE572BDC22F296FEA
uid [ultimate] daniel <danielnugrah@gmail.com>
sub rsa3072 2026-06-16 [E] [expires: 2028-06-15]

vboxuser@ubuntu: $
```

c. Export kunci publik Anda. Jalankan perintah ini:

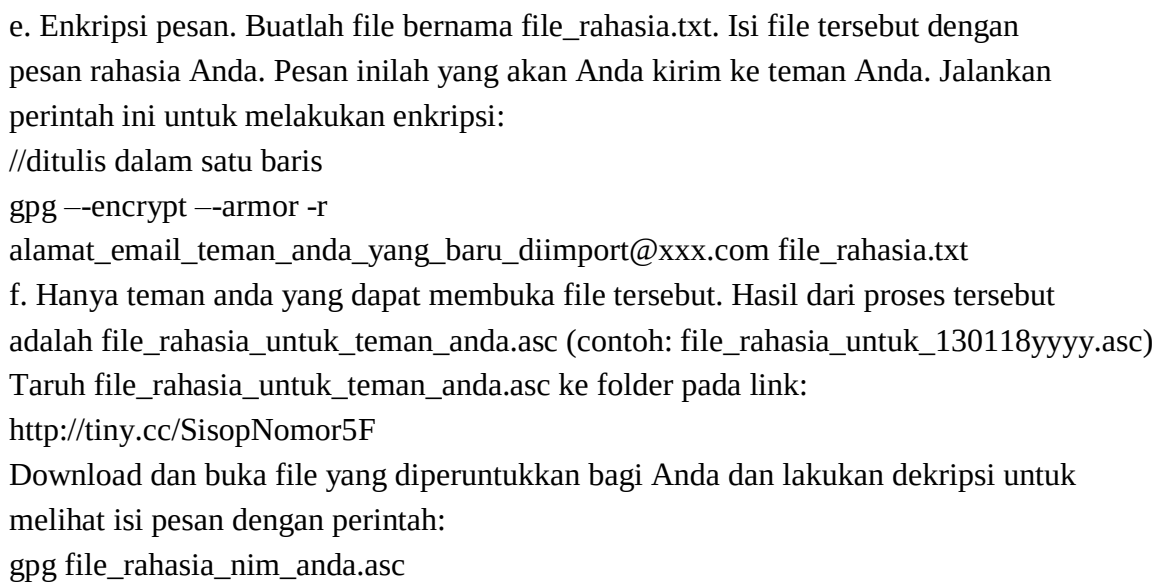
`gpg --armor --export nama@email.anda > mypublic_key.asc`

File `mypublic_key.asc` adalah file yang akan dibagikan kepada teman-teman Anda.

Teman Anda akan menggunakan kunci publik Anda jika ingin mengirim pesan rahasia kepada Anda. Hanya Anda yang dapat membaca pesan tersebut karena hanya mempunyai

- i. Rename mypublic_key.asc menjadi nim_anda.asc, Contoh: 130118xxxx.asc
- ii. Taruh file nim_anda.asc ke folder pada link berikut: <http://tiny.cc/SisopNomor5C>
- d. Mengimport kunci publik orang lain. Silakan download file nim_teman_sebelah_anda.asc dan jalankan perintah ini untuk mengimport (menambahkan kunci publik orang lain ke sistem Anda):

Contoh: “`gpg --import 130118yyyyy.asc`” Setelah mengimport kunci publik teman Anda, Anda dapat mengirimkan pesan kepada teman Anda secara terenkripsi. Untuk memastikan bahwa kunci publik telah diimport lakukan perintah `gpg --list-keys` dan nama teman Anda ada pada hasil perintah tersebut.



**Untuk e&f tidak bisa karena firework di dalam ubuntu error tidak bisa di buka :)
Jadi tidak bisa membuka link yang di berikan.**